A hand on strings

A mechanical hand toy stands open with five strings at the wrist. What inside a real hand works like it?

Bones in a row

Each finger has three small bones in a line, and the thumb has two. One whole hand holds 27 bones.

A string pulls

Forearm muscles pull long tendons, like strings. Tendons pull finger bones, and loops keep them close, like the toy's loops.

Where fingers bend

Joints are where bones meet. Finger joints fold like door hinges. The thumb's base joint lets it swing across the palm.

String to finger

Pull each toy string and match it to a finger. Then curl a real finger and feel the forearm move.

Opening again

Muscles only pull, never push. Tendons on the back of the hand pull fingers straight again. Toys may use elastic.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

How a Hand Works